RESEARCH REPORT · VITKI

Fully tested finding

Children inherited parts of their parents' names – parent and child share a name element almost twice as often as chance gives

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Timeline of the study: The material was frozen 28 October 2026 · The analysis plan was locked 29 October 2026 · The analysis was run 29 October 2026 · The addendum plan for the sensitivity analyses was locked and run 31 October 2026 · The exploratory shuffle within hundreds was run 31 October 2026

ABSTRACT

Background and aim. In the Germanic naming custom a child could receive an element of a parent's name, like Holmfast after Holmger. Onomastics describes the custom, but how often it shows on the runestones has not been measured against chance. The question here is whether parent and child, and siblings, share name elements more often than chance gives, and whether the custom weakened over time and with Christianity.

Material. 1,292 parent–child pairs from 714 stones and 1,066 sibling pairs from 488 stones, from the relations on the 2,321 Swedish Viking-Age runestones in the Scandinavian Runic-text Database (Rundata).

Method. A pre-registered study with five tests and Holm correction. The names are divided into first and second elements from the material itself, and the share of pairs sharing an element is compared with 9,999 shuffles of the names within the province. Internal replication in two halves and six sensitivity analyses under a locked addendum plan.

Results. Parent and child share a name element in 5.3 percent of the pairs, against 2.8 percent with random pairing, Holm-corrected $p \leq 0.0005$. It holds in both halves and in every sensitivity analysis, also without Uppland, with time held constant and with one pair per stone, and in an exploratory shuffle within the hundred (3.3 percent expected). Siblings share name elements in 6.7 percent against 2.9; that excess shrinks when the names are shuffled only within the hundred. No difference is shown over time (excess 3.1 against 2.6 percentage points, $p = 0.89$). On stones with Christian markers the excess is smaller (1.1 against 2.9 percentage points), but the difference is not clear ($p = 0.22$).

Conclusion. Children received elements of their parents' names, but it was a custom for a minority, about one pair in twenty. No weakening during the runestone period can be shown; on the Christian stones the excess is smaller, but the difference may be chance.

Keywords: *runestones, Viking Age, personal names, name elements, variation, name-giving, family, permutation test, pre-registration*

HIGHLIGHTS

- Children sometimes received an element of a parent's name, like Holmfast after Holmger, and it shows on the runestones more than chance gives.
- It was a possibility for some families, not a rule for all: about one pair in twenty.
- No weakening of the naming custom during the runestone period can be shown; on the Christian stones the excess is smaller, but that may be chance.

1. Introduction

Germanic personal names are often compounds of two elements, and onomastics has long described how the elements passed on within the family, through variation (an element recurs, Holmger and Holmfast), alliteration (the same initial sound) and repetition of the whole name (Janzén 1947). Lena Peterson (2007) has listed the names of the runestones and their elements. Birgit Sawyer (2000) read the runestones as expressions of family and inheritance.

What is missing is a measure of how often the name elements actually pass on on the runestones, compared with what chance gives when one allows for some elements being common in a district, and whether the custom changed when Christianity came. The aim here is to measure that.

2. Pre-registration: hypothesis and plan

The hypotheses, measures and rules of interpretation were written down in an analysis plan on 29 October 2026, before any name pairs had been compared. The plan has the SHA-256 checksum `74eb51136233aa89…` (see Data and code). Before locking only the number of relations was known and one example, U 135, where the son bears the father's name.

The plan tests five hypotheses with Holm correction: H1 that parent and child share name elements more often than chance gives, H2 that they alliterate more often, H3 that the excess is smaller on late stones, H4 that it is smaller on stones with Christian markers and H5 that siblings share name elements more often than chance gives. A hypothesis *holds* when Holm p is below 0.05 and the direction is the same with p below 0.05 in both halves.

The plan did not contain the project's fixed sensitivity analyses, which are built for associations between two traits on a stone. The corresponding steps for name pairs were therefore fixed in an addendum plan on 31 October 2026 (SHA-256 `9d0247a8c21ba9b7…`), before they were computed. One of the steps, without short elements, had been computed after the main run and was locked as a step in the addendum plan. A shuffle within the hundred (section 5.4) is exploratory: it was computed after the addendum plan, to test a weakness the plan did not address.

3. Material

- **The stones.** The 2,321 Swedish Viking-Age runestones in the Scandinavian Runic-text Database (Rundata), frozen for the analysis on 28 October 2026. Runestones only.
- **The relations** – father, mother, son, daughter, brother and sister – and the names are read from Rundata's translation and normalisation with a purpose-built rule set for persons and kin terms. It has not been checked against hand coding of name pairs; a related rule for the dead person's kin relation was right on 38 of 40 stones in a new sample. The pair is also counted when the sponsor bears the same name as the dead father, mother or son; otherwise the rules never make a person kin to themselves.
- **The pairs.** 1,292 parent–child pairs from 714 stones and 1,066 sibling pairs from 488 stones, each pair of names once per parish, so that a family with many stones, like Jarlabanke's, is not counted several times.
- **The name elements.** A name is divided into a first and a second element when both elements occur in at least three other names in the material; otherwise the whole name counts as one element. Two names *share an element* when they have an element in common or are the same name.
- **Time** by the stone's style group (Gräslund 2006): early stones RAK–Pr2, late Pr3–Pr5.

4. Method

Statistical terms are explained in the glossary, https://vitkirunor.se/en/forskningsrapporter/begrepp.

4.1 The measure. The share of pairs sharing a name element is compared with the share when the second name in each pair is shuffled at random among the pairs within the same province, 9,999 times. The shuffle keeps the province's stock of names, so that an element common in a district is not counted as inheritance. The test is one-sided: it asks only whether the pairs share elements more often than chance gives.

4.2 The other hypotheses. H2 as H1 with the same initial sound as the measure. H3 and H4 compare the excess per pair, observed minus expected, between late and early stones and between stones with and without Christian markers respectively. Those tests are two-sided: a difference in either direction counts.

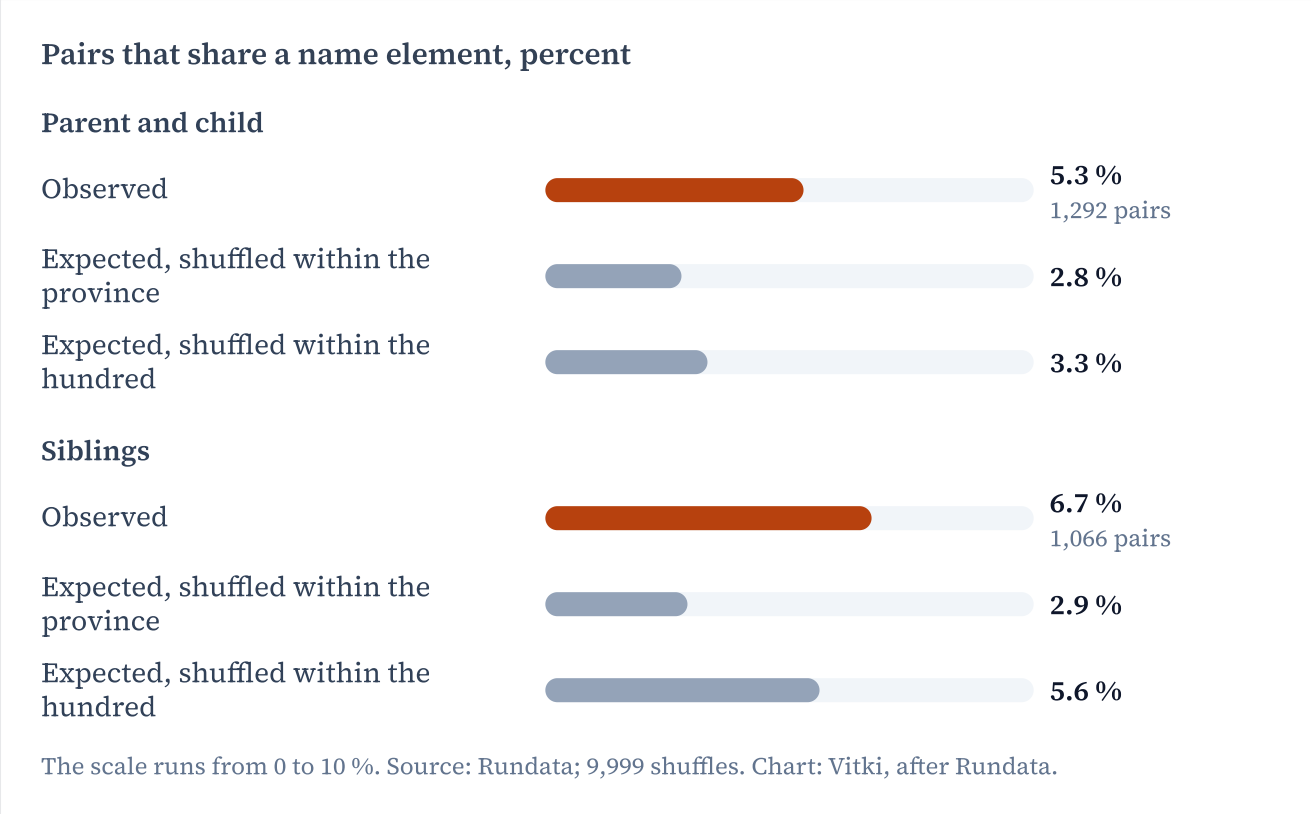
4.3 Correction and replication. Holm correction (Holm 1979) over the five tests. Each hypothesis is also tested in each half on its own.

4.4 The sensitivity analyses under the addendum plan, for H1 and H5: the halves; without one province at a time (provinces with at least 30 pairs; significance required only where at least two thirds of the pairs remain); time held constant, with the shuffle within province and period; one pair per stone; without short elements and the data-driven junk elements -ni, -bbi and -einn; and only shared elements, without pairs where both bear the same whole name. A step holds when the share is greater than the expected one and p is below 0.05.

4.5 Deviations from the plan. None. The addendum run gives the same shares as the main run; the halves' p-values differ in the last decimal because the random draws come in a different order.

5. Results

5.1 Parent and child (H1). 1,292 pairs: 5.3 against 2.8 percent, $p \leq 0.0001$ (the smallest possible with 9,999 shuffles), Holm-corrected $p \leq 0.0005$. Half A: 4.9 against 2.3 percent, $p = 0.0002$. Half B: 5.8 against 3.4 percent, $p = 0.0005$. H1 *holds*. In plain words: about one parent–child pair in twenty shares a name element, against one in thirty-five if the names were distributed at random within the same province. The elements most often passed on are *-björn*, *Tor-*, *-sten*, *-fast*, *Sig-* and *-mund*, for example Holmger and Holmfast (U 289), Ingevald and Ingemund (U 296) and Gudmar and Gudbjörn (Sö 164). On 6 stones the son bears his father's whole name, as Östen after Östen (U 135).



5.2 Siblings, alliteration, time and Christianity (H5, H2, H3, H4). Siblings: 1,066 pairs, 6.7 against 2.9 percent, $p \leq 0.0001$, Holm-corrected $p \leq 0.0005$, and in both halves ($p \leq 0.0001$ and $p \leq 0.0001$); H5 *holds*. Alliteration between parent and child: 14.9 against 12.5 percent, Holm-corrected $p = 0.013$, but only in half B; an indication. Over time: the excess is 3.1 percentage points on late stones and 2.6 on early ones, $p = 0.89$; no difference is shown. Christian markers: the excess is 1.1 percentage points on stones with Christian markers against 2.9 on the others, $p = 0.22$. It is smaller in both halves (−1.0 and −2.5 percentage points), but not clearly so in either; the difference is not shown. Mother and child: only 27 pairs, too few to say anything.

5.3 The sensitivity analyses. All hold, for both parent–child and siblings:

- Time held constant: parent–child 5.3 against 2.8 percent, $p \leq 0.0001$; siblings 6.7 against 3.3 percent, $p \leq 0.0001$.
- One pair per stone: parent–child (714 pairs) 5.6 against 2.8 percent, $p \leq 0.0001$; siblings (488 pairs) 7.2 against 2.5 percent, $p \leq 0.0001$.
- Without short elements: parent–child 4.5 against 2.5 percent, $p \leq 0.0001$; siblings 6.0 against 2.7 percent, $p \leq 0.0001$.
- Only shared elements, without the same whole name: parent–child 4.9 against 2.3 percent, $p \leq 0.0001$; siblings 6.7 against 2.3 percent, $p \leq 0.0001$.
- Without one province at a time: without Uppland 534 parent–child pairs remain, fewer than two thirds, and the direction holds (4.5 against 2.8 percent, $p = 0.011$); without each of the other provinces with at least 30 pairs $p \leq 0.0001$. The siblings likewise.

5.4 Shuffling within the hundred (exploratory). Families in the same district share a local stock of names, and shuffling within the province may therefore overstate the inheritance. If the names are shuffled only within the hundred, the expected share for parent–child becomes 3.3 percent against the observed 5.3, $p \leq 0.0001$, in 113 hundreds: the finding remains. For siblings the expected share becomes 5.6 percent against 6.7, $p = 0.053$: the excess shrinks and is no longer clear. That shuffle may however be biased upwards for siblings, since a stone with several siblings gives many pairs and in a small hundred the random partner often becomes another sibling in the same family; this has not been tested.

6. Discussion

What the finding says. Children received elements of their parents' names more often than chance gives, also with the district's stock of names held constant, and regardless of time, province and how many names a stone has. But the custom concerned a minority: about one pair in twenty, against one in thirty-five expected. It was a possibility in name-giving, not a rule. Over time no change is seen. On stones with Christian markers the excess is smaller, but the difference is not clear: the material can show neither that the custom weakened in the Christian families nor that it stayed unchanged.

Against earlier research. The result gives a measure to the naming custom onomastics has described (Janzén 1947) and shows that it is visible in the runestones' relations, not only in single examples. Whether the custom was weaker in the families that marked their Christian faith the material cannot decide; the direction points that way, but the difference lies within the range of chance.

Weaknesses, in order of weight.

1. **The district's stock of names.** Neighbours share names, and part of the excess may be local fashion. For parent–child the finding remains within the hundred; for siblings it shrinks (5.4).
2. **The division into elements is crude.** It rests on the material itself and gives some junk elements. Without short and junk elements the finding remains, but hypocoristic forms like Holmi count as single-element names and may hide inheritance.

3. **The relations are read by rules.** The name pairs have not been checked against hand coding (section 3). A misread relation creates a false pair or loses a real one.
4. **The children are adults.** The sponsor is usually an adult, so the naming lies 30–50 years before the stone's style group. The comparison over time thus concerns name-giving around 950–1100, with early stones RAK–Pr2 and late ones Pr3–Pr5, not the stones themselves.

What would overturn the finding. If the excess disappeared with a shuffle within the parish, once the material is large enough for it, or if a hand coding of the elements gave the same share for parent–child as for random pairs.

7. Conclusion

On the runestones parent and child share a name element in 5.3 percent of the pairs, against 2.8 percent with random pairing, and it withstands every sensitivity analysis. Children inherited elements of their parents' names, but only in a minority of families. No weakening during the runestone period can be shown.

Data and code

The analysis was run in Vitki 3.0.0, an open research tool for runestones (https://vitkirunor.se/en): the main analysis at commit `806b915` (2026-10-29), the sensitivity analyses at commit `9e158b6` and the shuffle within hundreds at commit `d77c99f` (both 2026-10-31) in https://github.com/Precatio/runes, where the result files were saved. All code, the analysis plans and the result files are openly available there under GPL-3.0.

- The material: corpus `3f58e9339191761e`, Rundata's Swedish Viking-Age runestones as frozen on 2026-10-28 (`data/cache/r/corpus/3f58e9339191761e/corpus.csv`, built from `data/rundata.json`).
- The analysis plans: `analysplaner/2026-10-29-namngivning-i-familjen.md` and `analysplaner/2026-10-31-namnled-trappan.md`, with their checksums in the `.sha256` files beside them.
- The analysis: `utdata/namngivning/analys.py` (results `namngivning.json`, the pairs in `par.csv`); the sensitivity analyses `trappa.py` (`trappan.json`); the exploratory shuffle within hundreds `harad.py` (`harad.json`).
- To rerun the analysis, from the project folder: `.venv/bin/python utdata/namngivning/analys.py`, `.venv/bin/python utdata/namngivning/trappa.py` and `.venv/bin/python utdata/namngivning/harad.py`.
- The names and relations: `src/entities.py`; the check of the dead person's kin relation: `src/people.py` against `data/people_check.json` (`utdata/namngivning/rapport_kontroll.py`).
- The report is built by `utdata/blogg/build/build_rapport_namnled.py`, which reads every number from the result files and stops if anything disagrees; the builder and the check file as reviewed before archiving are in commit `6eae0af`.

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Change log

- **Version 2, 2026-11-05.** Corrections before the permanent archiving in Zenodo, after a review of the report against the result files and the code. The main results (parent–child and siblings) and the level are unchanged.
- *Christian markers (H4).* Version 1 said that the custom was as strong on stones with Christian markers and that it did not weaken. The excess is smaller there (1.1 against 2.9 percentage points, $p = 0.22$), in the same direction in both halves, but not clearly so. The abstract, sections 5.2, 6 and 7 and the essentials box now say that a weakening cannot be shown and that the custom is somewhat weaker on the Christian stones, but not clearly so.
- *The check of the relations.* Version 1 stated that the relations were right on 38 of 40 stones and that the errors were about one in twenty. That check concerned a related rule for the dead person's kin relation, not the name pairs, which have not been checked against hand coding. Section 3 and weakness 3 are corrected.
- *p-values at the floor.* $p = 0.0001$ is the smallest possible with 9,999 shuffles and is now written $p \leq 0.0001$ (Holm-corrected $p \leq 0.0005$).
- *Siblings within the hundred.* That the shuffle within the hundred is biased upwards for siblings was an assumption; it now stands as a possibility that has not been tested.
- *The time of the naming.* The comparison over time concerns name-giving around 950–1100, not around the year 1000.
- *Language and form.* The essentials box says that children *sometimes*, not often, received an element of a parent's name; the sentence on sponsors with the same name as the dead is clarified; the number of stones is given in Material; the timeline has the freeze date; Data and code gives the commits and commands to rerun the analysis instead of a future DOI; Rundata is cited with version and licences; a link to the glossary, an explanation of one-sided and two-sided tests and the chart's source line are added.
- **Version 3, 2026-11-05.** Corrections after a second, independent reading of the changes in version 2. The main results (parent–child and siblings) and the level are unchanged.

- *Christian markers (H4).* Version 2 said that the custom is "somewhat weaker" on the Christian stones, "but not clearly so". That read the estimate as a property of the custom, although the test by the plan showed no difference (p = 0.22, and not clear in either half). The lede, the abstract's conclusion and the essentials box now say that the excess is smaller but that the difference may be chance.
- *Data and code.* The builder and the check file as reviewed before archiving are named with commit `6eae0af`.
- *What version 2 also changed without logging it.* The timeline also gained the date of the exploratory shuffle within hundreds (2026-10-31).

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